Quantum Optics

Maxwell's equations (vacuum)

$$\begin{cases} \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \times \vec{B} = \mu_0 \left(\varepsilon_0 \dfrac{\partial \vec{E}}{\partial t} + \vec{J} \right) \\ \nabla \cdot \vec{E} = \rho / \varepsilon_0 \\ \nabla \cdot \vec{B} = 0 \end{cases}$$

$[(\vec{r};t)$ omitted$]$

Potentials

$\vec{A}, \ \phi, \ \nabla \times \nabla \phi \equiv 0$

$$\begin{cases} \vec{B} = \nabla \times \vec{A} \\ \vec{E} = -\nabla \phi - \dfrac{\partial \vec{A}}{\partial t} \end{cases}$$

Field Equations

$$\begin{cases} \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} + \dfrac{1}{c^2} \dfrac{\partial \nabla \phi}{\partial t} + \dfrac{1}{c^2} \dfrac{\partial^2 \vec{A}}{\partial t^2} = \mu_0 \vec{J} \\ -\varepsilon_0 \nabla^2 \phi - \varepsilon_0 \nabla \cdot \dfrac{\partial \vec{A}}{\partial t} = \rho \end{cases}$$

Coulomb gauge

$\nabla \cdot \vec{A} = 0$

$$\begin{cases} -\nabla^2 \vec{A} + \dfrac{1}{c^2} \dfrac{\partial \nabla \phi}{\partial t} + \dfrac{1}{c^2} \dfrac{\partial^2 \vec{A}}{\partial t^2} = \mu_0 \vec{J} \\ -\nabla^2 \phi = \rho / \varepsilon_0 \rightarrow \text{Poisson equation} \end{cases}$$

$\rho(\vec{r};t) \rightarrow \phi(\vec{r};t)$ informs <u>istantaneously</u> the variation of $\rho(\vec{r},t)$

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Helmoltz theorem

Any vector field $\vec{F}$ can be decomposed into a curl free field $\vec{F}_L$ and a divergence free field $\vec{F}_T$

$\vec{F}$ → Long. field $\quad \vec{F}_L: \quad \nabla \times \vec{F}_L \equiv 0$

$\vec{F}$ → Transv. field $\quad \vec{F}_T: \quad \nabla \cdot \vec{F}_T \equiv 0$

The current density can be decomposed in Long. and Transv. comp.

$\vec{J} = \vec{J}_L + \vec{J}_T \qquad \vec{A}$ in Coulomb's gauge is only transverse $(\nabla \cdot \vec{A} = 0)$

$\vec{E}_T$ and $\vec{B}$ are also transverse and derive from $\vec{A}$ only

$\vec{E}_T = -\dfrac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \nabla \times \vec{A}, \quad \vec{E}_L$ derives from $\phi: \vec{E}_L = -\nabla \phi$

Field equation

$$\begin{cases} \underset{(T)}{-\nabla^2 \vec{A}_T} + \underset{(L)}{1/c_2 \dfrac{\partial \nabla \phi}{\partial t}} + \underset{(T)}{1/c^2 \dfrac{\partial^2 \vec{A}_T}{\partial t^2}} = \underset{(T)}{\mu_0 \vec{J}_T} + \underset{(L)}{\mu_0 \vec{J}_L} \end{cases}$$

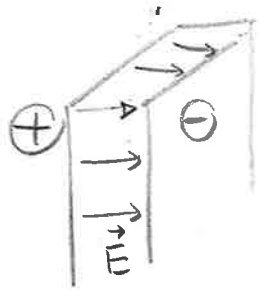
$$\begin{cases} -\nabla^2 \vec{A}_T + 1/c^2 \dfrac{\partial^2 \vec{A}_T}{\partial t^2} = \vec{J}_T \\ 1/c^2 \dfrac{\partial \nabla \phi}{\partial t} = \mu_0 \vec{J}_L \end{cases}$$

Non homogeneous D'Alembert eq.

$\rightarrow \quad \vec{J}_L = -\varepsilon_0 \dfrac{\partial \vec{E}_L}{\partial t} \qquad \vec{J}_L = -\vec{J}_D$

$\left[c^2 = \dfrac{1}{\varepsilon_0 \mu_0} \right]$

$\vec{J}_L$ change the charge distribution



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Homogeneous field equation in Coulomb gauge

Solution in a quantization cubic cavity of size L

$$-\nabla^2 \vec{A} + \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} = 0 \qquad \vec{J}_T = 0$$

Periodic boundary condition

→ Field mode at the boundaries

The solution is a superposition of modes (Plane waves)

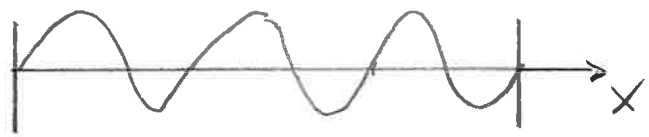
$$\begin{cases} \vec{A}(\vec{r},t) = \sum_{\vec{k}} \sum_{\lambda=1,2} \vec{e}_{\vec{k}\lambda} A_{\vec{k},\lambda}(\vec{r},t) \\ A_{\vec{k},\lambda}(\vec{r},t) = A_{\vec{k}\lambda} \exp(-i\omega_k t + i\vec{k}\cdot\vec{r}) + c.c. \end{cases}$$

$\vec{k}$ = wavevector

$$k_x = \frac{2\pi n_x}{L} \qquad n_x = 0, \pm 1, \pm 2 \ldots$$

$$k_y = \frac{2\pi n_y}{L} \qquad n_y = 0, \pm 1, \pm 2 \ldots$$

$$k_z = \frac{2\pi n_z}{L} \qquad n_z = 0, \pm 1, \pm 2$$



Modes are independent from each others and behave as harmonic oscillators

$\vec{E}_T$ and $\vec{B}$ fields

$$\vec{E}_T = -\frac{\partial \vec{A}}{\partial t} = \sum_{\vec{k}} \sum_{\lambda} \vec{e}_{\vec{k}\lambda} E_{\vec{k}\lambda}(\vec{r},t) \leftarrow \text{modes}$$

$$E_{k\lambda}(\vec{r},t) = i\omega_{k\lambda}\{A_{\vec{k}\lambda} \exp(-i\omega_k t + i\vec{k}\cdot\vec{r}) + c.c.\}$$

$$\vec{B} = \nabla \times \vec{A} = \sum_{k} \sum_{\lambda} \vec{k} \times \vec{e}_{\vec{k}\lambda} B_{\vec{k},\lambda}(r,t)$$

$$B_{\vec{k}\lambda}(\vec{r},t) = i\vec{k}\{A_{\vec{k}\lambda} \exp(-i\omega_k t + i\vec{k}\cdot\vec{r}) + cc\}$$

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Energy in the quantization cavity

$$\mathcal{E} = \frac{1}{2}\int_{cavity}\left[\varepsilon_0 E_T(\vec{r},t)\cdot E_T(\vec{r},t) + \mu_0^{-1} B(\vec{r},t)\cdot B(\vec{r},t)\right] dV$$

Modes are independent from each other => the energy of the field is the sum of the energy of the modes (modes are orthogonal)

$$\mathcal{E} = \sum_{\vec{k}}\sum_{\lambda} \mathcal{E}_{\vec{k},\lambda} \qquad \mathcal{E}_{\vec{k}\lambda} = \varepsilon_0 V \omega_k^2 \left(A_{\vec{k},\lambda} A^*_{\vec{k},\lambda} + A^*_{\vec{k}\lambda}\cdot A_{\vec{k}\lambda}\right) = 2\varepsilon_0 V \omega_k^2 \left|A_{\vec{k}\lambda}\right|^2$$

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